

wait until vs repeat until

Choosing the right sensor pattern in Python

Which code **matches** the robot's goal?

The robot should move forward until the force sensor is pressed, then stop.

Option A

```
wait until (is pressed)  
move forward
```

Option B

```
repeat until (is pressed):  
    move forward()
```

Option C

```
move forward  
wait until(is pressed)
```

Option D

```
while not is pressed():  
    move forward()
```

Which code **matches** the robot's goal?

The robot should move forward until the force sensor is pressed, then stop.

Option A
wait until (is pressed)
move forward

Option B
repeat until (is pressed):
 move forward()

Option C
move forward
wait until(is pressed)
Correct Answer

Option D
while not is pressed():
 move forward()

Correct answer: This is a wait until pattern because the robot moves first and then waits until the sensor becomes true.

Which code **matches** the robot's goal?

The robot should move backward until the distance sensor is less than 20 cm, then stop.

Option A

```
move backward  
wait until(is_near(20))
```

Option B

```
while not is_near(20):  
    move backward
```

Option C

```
wait until(is_near(20))  
move backward
```

Option D

```
while is_near(20):  
    move backward
```

Which code **matches** the robot's goal?

The robot should move backward until the distance sensor is less than 20 cm, then stop.

Option A

```
move backward  
wait until(is_near(20))
```

Option B

```
while not is_near(20):  
    move backward
```

Correct Answer

Option C

```
wait until(is_near(20))  
move backward
```

Option D

```
while is_near(20):  
    move backward
```

Correct answer: This is a repeat until pattern because the robot keeps moving backward while the distance is not close enough. Stops when the condition becomes true.

Which code matches the robot's goal?

The robot should keep turning left until the color sensor sees blue, then stop.

Option A
turn left()
Wait until(is blue)

Option B
while not is blue():
 turn left()

Option C
Turn left()
while not is blue():
 turn left()

Option D
while is blue():
 turn left()

Which code matches the robot's goal?

The robot should keep turning left until the color sensor sees blue, then stop.

Option A
turn left()
Wait until(is blue)

Option B
while not is blue():
 turn left()
Correct answer

Option C
Turn left()
while not is blue():
 turn left()

Option D
while is blue():
 turn left()

Correct Answer: This is a repeat until pattern because the robot keeps turning while the color is not blue. Stops when the condition becomes true.

Which code **matches** the robot's goal?

The robot should keep moving forward while the distance sensor is more than 8 cm away. Stops when it gets close.

Option A

```
while is_near(8):  
    start moving forward  
stop moving
```

Option B

```
start moving forward  
wait until is_near(8)  
stop moving
```

Option C

```
while not is_near(8):  
    start moving forward  
stop moving
```

Option D

```
wait until not is_near(8)  
move forward()
```

Which code **matches** the robot's goal?

The robot should keep moving forward while the distance sensor is more than 8 cm away. Stops when it gets close.

Option A

```
while is_near(8):  
    start moving forward  
stop moving
```

Option B

```
start moving forward  
wait until is_near(8)  
stop moving
```

Option C

```
while not is_near(8):  
    start moving forward  
stop moving  
Correct Answer
```

Option D

```
wait until not is_near(8)  
move forward()
```

Correct answer: This is a repeat until pattern because the robot keeps moving while the distance is still greater than 8 cm. When the robot gets close enough, the condition becomes false and the loop stops.

Which code **matches** the robot's goal?

The robot should keep checking the color sensor until it sees red, then stop.

Option A

```
while not is red():  
    move forward()
```

Option B

```
move forward()  
wait until is red
```

Option C

```
while is red():  
    move forward()
```

Option D

```
wait until is not red  
move forward()
```

Which code **matches** the robot's goal?

The robot should keep checking the color sensor until it sees red, then stop.

Option A

```
while not is red():  
    move forward()
```

Correct Answer

Option B

```
move forward()  
wait until is red
```

Option C

```
while is red():  
    move forward()
```

Option D

```
wait until is not red  
move forward()
```

Correct answer: This is a repeat until pattern because the robot keeps moving while the sensor is not red. When the robot finally sees red, the condition becomes true and the loop stops.